CHAPTER

19

Introduction to UX Evaluation: Concepts and Terminology

Highlights

- Types of UX evaluation data
  - Quantitative vs. qualitative data
  - Objective vs. subjective data
- Types of UX evaluation methods by goal
  - Formative vs. summative evaluation
  - Informal vs. formal summative evaluation
  - Analytic vs. empirical UX evaluation methods
- Concerns about UX evaluation methods
- UX evaluation within the pyramid of human needs
- UX evaluation goals and constraints determine method choices

19.1 INTRODUCTION

19.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 19.1). We begin Part 5 with an introduction to UX evaluation concepts and terminology. The how-to details for doing UX evaluation are in the rest of the Part 5 chapters.

The design chapters were about getting the right design. These evaluation chapters are about verifying, validating, and refining that UX design to ensure the design is correct. Evaluation methods were a central focus of early usability and human–computer interaction (HCI) research and publication, so there are many methods, and they account for more literature than other equally deserving topics. We will do our best to boil down the approaches you are likely to encounter or need. Evaluating UX activities includes:

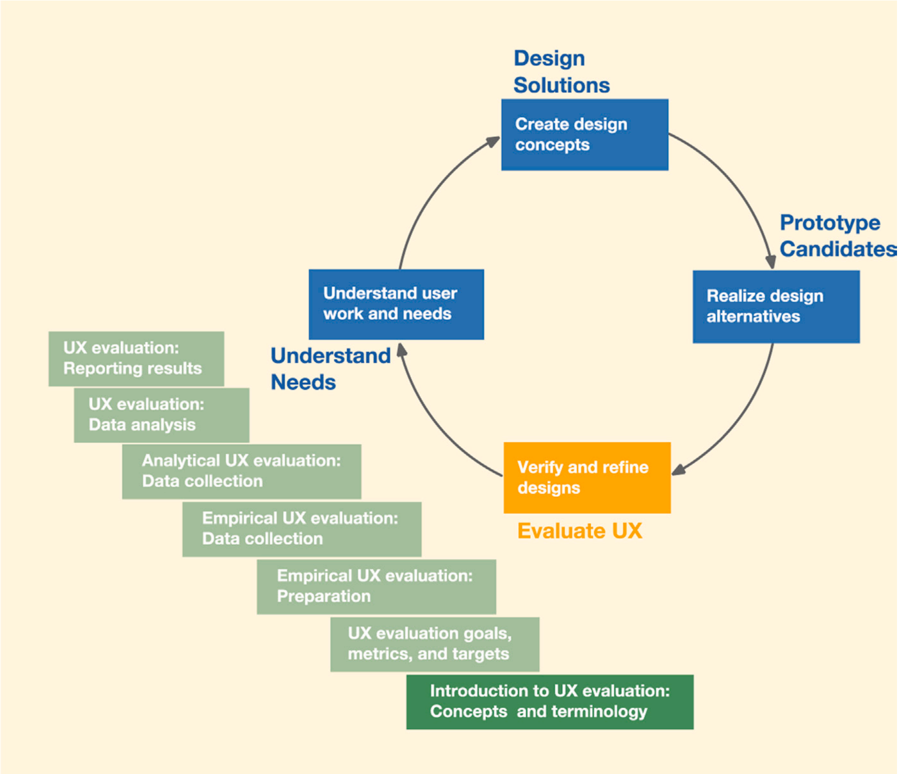

Fig. 19.1

You are here in the upfront chapter on UX evaluation concepts within the evaluate UX activity in the context of the overall Wheel UX lifecycle.

- Plan and prepare by clearly identifying the evaluation goals (see Section 21.3, Chapter 20)
- Collect evaluation data by evaluating designs with empirical methods (see Chapter 22) or analytic methods (see Chapter 23)
- Analyze evaluation data to identify causes of UX problems found (see Chapter 24)
- Report results to all stakeholders about UX problems discovered along with recommendations for fixes (see Chapter 25)

Methods abound for doing the activities of UX evaluation, from lightweight and rapid to thorough and laborious. There are also different methods for evaluating the different components of UX, such as usability, usefulness, emotional impact, and meaningfulness.

19.1.2 User Testing? No!

User testing is a term we see or hear expressed by non-UX people; however, such a framing can have the negative outcome of making users feel that they are being tested. No user will like the idea of being tested and, thereby, possibly made to look ridiculous. Instead, the goal is to have real users help us test the design.

So, we refer to this as UX evaluation, and we call our users participants in this context, clearly recognizing their role as partners in uncovering UX problems.

19.2 TYPES OF UX EVALUATION DATA

Procedurally, you will not have to worry about the different types of data. The methods will lead you to the right choices. If you want to be a true UX practitioner, beyond just being a UX technician, you will want to understand the terminology and the nature of what you are doing at a deeper level.

19.2.1 Quantitative vs. Qualitative Data

19.2.1.1 Quantitative data

Quantitative data are numeric, usually from measurements, and are mostly used to assess a level of achievement of UX quality. Generally, quantitative data are assessment data, which help you determine how well you are doing.

You would choose to use quantitative data when you need to do informal summative evaluation (Section 19.4.1) with the goal of helping the team assess UX achievements and monitor convergence toward specified levels set in the UX targets (see Chapter 20). In this kind of assessment, an important characteristic of quantitative measurements is that they can be compared. For example, you can compare the time to complete a given task, before and after a design change, to get an informal idea of which is better.

You will also use quantitative data if you need to do formal summative evaluation in A/B testing (Section 19.4.2.1; also see Section 22.8.4) and large-scale customer satisfaction (CSAT) studies (Section 19.4.2.2; also see Section 22.2.2.3).

Example: Quantitative UX Evaluation Data

A good example of quantitative data is numeric measurement of user performance, such as the time to complete a given task or the number of errors made in the process. A questionnaire can also yield quantitative data if choices are linked to numeric values.

19.2.1.2 Qualitative data

Qualitative data are non-numeric descriptive data used to describe UX problems or issues observed or experienced during usage or reported in satisfaction surveys. An example of this latter type is when we ask users of a survey to describe their reasons for the rating they provided for a particular question. The responses to such open-ended questions in a survey constitute qualitative data. Generally, qualitative data are diagnostic data (i.e., keys to identifying UX problems and their causes within formative evaluation).

You would choose qualitative data in the form of UX problem descriptions when you are doing formative evaluation (Sections 19.3.1 and 19.5). Qualitative data are key to finding and fixing UX problems, and therefore key to improving usability and other UX qualities. In this regard, qualitative data are arguably one of the most important kinds of UX evaluation data.

Of all the methods for collecting and using qualitative UX evaluation data, critical incident identification is one of the most important; critical incident identification (see Section 22.3.1) is the workhorse of empirical qualitative data collection techniques. A critical incident is a usage event that reveals something significant about the quality of the user experience. Critical incidents are usually observed by the evaluator, perhaps with help from the participant, usually during task performance.

User think-aloud qualitative data collection techniques (see Section 22.3.2) are a close second place in terms of importance. Also called verbal protocol, the think-aloud technique is a qualitative empirical data collection technique, in which user participants are encouraged to verbally express their thoughts about the interaction experience as they perform tasks and exercise a UX design. The technique is simple to use for both UX evaluator and participant. It is most useful during empirical evaluation of user task performance, but it is also useful when a participant walks through a prototype or helps you with a UX inspection.

You may choose the critical incident identification method of qualitative data collection if you are observing user participants performing tasks. You may choose the think-aloud method of qualitative data collection if your formative evaluation setup allows for users to think and talk while they perform tasks. See Section 22.3.2 for more about user think-aloud data collection methods for gathering qualitative UX data.

Example: Qualitative UX Evaluation Data

The following examples are UX problem descriptions of qualitative evaluation data:

> The user of an online shopping website knows he had to click on a button that said something to the effect of "Submit order," but he could not find it. It turns out that there is a "Proceed to checkout" button, but it is at the very end of the ordering information, scrolled down out of sight.

There are two buttons in a screen design that are very similar, the "Find" button and the "Search" button. The developers say there are subtle but important differences, but users are confused about which one to use.

19.2.2 Objective vs. Subjective Data

In a separate dimension, we have objective and subjective data. Practically, the two dimensions are orthogonal, so both objective and subjective data can be either qualitative or quantitative.

19.2.2.1 Objective UX data

Objective data are acquired from direct observation or measurement, often of user task performance, by either the UX evaluator or participant. By definition, objective data are always associated with empirical methods.

Example: Objective UX Data

Data, such as a measurement of the time it takes for a participant to complete a given task or the number of errors a user commits as part of that task, are objective because they are based on direct observation.

19.2.2.2 Subjective UX data

Subjective data represent opinions or judgments of either the UX evaluator or the participant, often concerning the user experience and satisfaction with the design. For example, user questionnaires (see Section 22.2.2) can yield data that are quantitative (data on numerical scales), qualitative (data from open-ended responses to questions), and subjective (data based on opinions of users).

Example: Subjective UX Data

Questionnaire scores to rate perceived user satisfaction with a particular feature of a product on a scale of choices are subjective data because they depend on opinions (of those who complete the questionnaire).

19.3 TYPES OF UX EVALUATION METHODS BY GOAL: FORMATIVE VS. SUMMATIVE

One useful way to differentiate UX evaluation methods is by evaluation goals. Fig. 19.2 shows the structure of our discussion of types of UX evaluation methods.

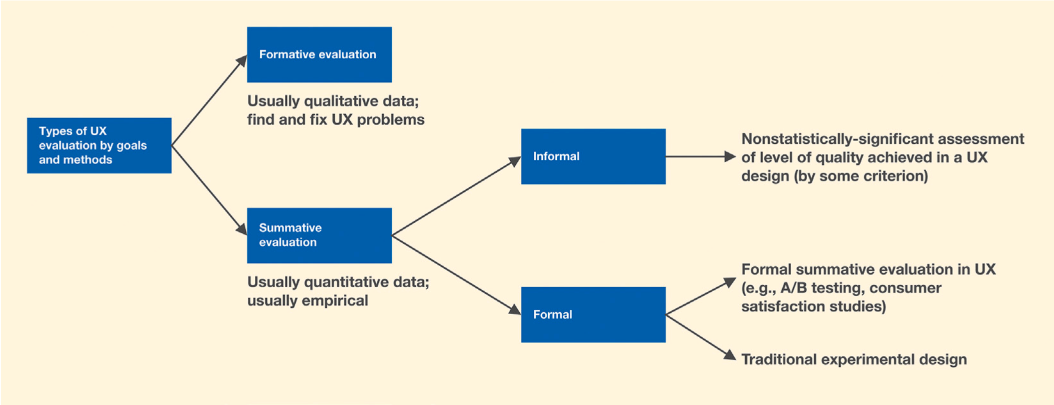

Fig. 19.2
Types of UX evaluation.

19.3.1 Formative UX Evaluation

The goal of formative UX evaluation is to form a design and is therefore primarily diagnostic (i.e., to find and fix UX problems to improve the design). Formative evaluation is the most emphasized and most used type of evaluation in practice. Using a low- or medium-fidelity prototype to get feedback from potential users on potential user experience problems is an example of formative evaluation.

19.3.2 Summative UX Evaluation

The goal of summative UX evaluation is to sum up a design. Summative UX evaluation is usually an empirical process (Section 19.5.1) in which we usually gather quantitative data to assess the level of UX quality achieved in a design. Summative evaluation is a comparative assessment, so the goal includes comparing the current design with either a previous version or a different design to determine which is better with respect to specific criteria, such as user performance with the design, expert analysis of the design, or opinions of users (e.g., via a questionnaire).

Formative and summative evaluation can be used together in close combination. For example, the same session of observing users perform a task could yield qualitative data for the formative component and quantitative data for an informal summative component.

19.4 TYPES OF SUMMATIVE EVALUATION METHODS BY PROCESS: INFORMAL VS. FORMAL

19.4.1 Informal Summative Evaluation

Informal summative evaluation is used to sum up or assess the quality and effectiveness of a UX design using informal metrics, which include quantitative data comparisons using non-statistically significant descriptive statistics, such as averages.

Often used as a close partner to formative evaluation, informal summative evaluation helps you judge progress of the UX design process and convergence of a design toward UX performance targets, using techniques described in Chapter 20 on UX evaluation goals, metrics, and targets. As a tool to guide and manage the overall process of UX evaluation, informal summative evaluation helps practitioners know when they can stop iterating the design (see Section 24.2.4).

19.4.2 Formal Summative Evaluation

Formal summative evaluation customarily refers to traditional experimental design (Keppel & Wickens, 2004; Winer et al., 1991). Formal summative evaluation requires rigorous methods for statistical tests to resolve narrow questions, such as which version of a design yields better effectiveness or a better quality UX experience with respect to a narrow criterion (e.g., comparing two interaction techniques to select a 12-inch object from a distance of 4 feet inside a virtual or augmented reality application).

Because it is costly and usually does not add value to UX practice, this kind of formal summative evaluation is rarely if ever used within modern UX practice and is beyond the scope of this book. Exceptions are A/B testing and CSAT studies (lower righthand corner of Fig. 19.2 and discussed in next sections), which are special cases of formal summative evaluation used in UX practice.

19.4.2.1 A/B testing in UX

A/B testing (see Section 22.8.4) is an effective and economical way to assess the impact of design variations on various criteria, such as user performance, by determining which version of a feature in your system is best. A/B testing requires that the system being evaluated be instrumented (e.g., by embedding measuring and tracking probes to capture fine-grained user actions and system responses in a log that can be analyzed). Large websites with high volumes of user traffic are an excellent source of this kind of data because of the ease with which they can be instrumented and the large amounts of data that can be generated. The large amounts of data combined with rigorous statistical methods put A/B testing in the formal summative category.

19.4.2.2 CSAT studies to measure customer satisfaction

The CSAT study is a formal summative evaluation method, in which users evaluate specific capabilities of a product or the product as a whole (see Section

22.2.2.3) through a survey. The CSAT study can have a qualitative component that provides open-ended qualitative feedback (positive or negative) on the product that falls in the formative evaluation category and is used to improve the design. CSAT studies also have a component that sums up a quantitative metric (survey score) that typically is statistically significant when the survey population is large enough and formal statistical analysis is used.

19.5 TYPES OF FORMATIVE UX EVALUATION METHODS BY PROCESS: EMPIRICAL VS. ANALYTIC

It is also useful to differentiate UX evaluation methods by process-related differences.

19.5.1 Empirical Formative UX Evaluation

The goal of formative UX evaluation methods is to find and fix UX problems. Differences, however, lie in how the UX problems are found. Empirical methods for finding UX problems yield UX problem data directly observed and/or measured. Empirical methods derive knowledge from actual experience (of using the system, for example) rather than from theory.

Informally, you could say that empirical data reflect something that happened or something that people did. The bottom line is that empirical UX evaluation depends on data observed in the performance of real users or user participants, data reflecting real user usage experience.

As an example, if a user participant is performing a task and a UX evaluator is measuring the time to perform the tasks, this is direct (firsthand) and observational, so it is empirical. A questionnaire administered to a user participant after task performance is also empirical because it is a firsthand report (direct) on usage experience (experiential).

If you want to evaluate a UX design based on actual user performance on representative tasks, then you will be doing empirical evaluation and collecting empirical data.

19.5.2 Analytic Formative Evaluation

The goal of analytic formative evaluation is still to find and fix UX problems. In contrast to empirical methods, analytic methods do not employ users or user proxies as experimental participants; instead, analytic methods are based on examining inherent attributes of the design. An example is a UX expert going through a prototype and looking for potential UX problems. Analytic formative methods yield subjective data. Analytic UX evaluation methods (see

Chapter 23) were developed as faster and less expensive methods to produce approximations to, or predictors of, empirical results.

If you want to evaluate a UX design based on inspection and expert opinions of UX problems and their causes, you will be doing analytic evaluation and collecting analytic data.

19.5.3 Payoff and Intrinsic Methods

Empirical methods are sometimes called payoff methods (Carroll, Singley, & Rosson, 1992; Scriven, 1967) because they are based on how a design or design change pays off in real observable usage. Analytic methods are sometimes called intrinsic methods because they are based on analyzing intrinsic characteristics of the design. Section 19.7.1 provides an analogy (of evaluating an axe) to explain the difference between empirical and analytic evaluation.

19.5.4 UX Evaluation Within the Pyramid of Human Needs

As a reminder, the UX pyramid of human needs has three layers (Fig. 19.3). Almost all the data collection methods and techniques in this chapter can be used to conduct UX evaluation within any layer of the needs pyramid, at any scope, and at any level of rigor.

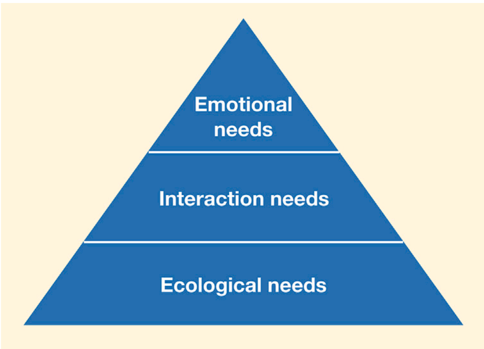

Fig. 19.3

Pyramid of human needs that are the purpose of UX design (from Fig. 11.5).

19.5.5 UX Evaluation Within the Ecological Level of the Pyramid of Needs

As usual, the early funnel (see Section 4.4.4) is best for evaluating in the ecological layer, where you have a large scope to embrace the product or system ecology and its conceptual design.

The system ecology is about the operation of the broader system and the various devices and contexts it encompasses. Issues with its design tend to be high level in nature and generally abstract. Therefore evaluating the conceptual design of an ecology requires measuring instruments that bring into focus the themes and structures of how the overarching system works.

Two main goals for evaluating the design of an ecology are as follows:

- Ascertain whether users understand how the ecology is structured:
  - Evaluate whether the designers' mental model is communicated clearly through the conceptual design.
  - Evaluate whether users are able to form a clear mental model as a result.
- Ensure the users are able to get work done in the new ecology. Evaluate whether the conceptual design of the ecology is appropriate for their work context.

Evaluate user understanding. Qualitative data are the most effective in formative evaluation of a system. After participants get familiar with the various devices and their capabilities, ask them to articulate how the overall system works. Asking them to explain how the system works to a new colleague is an effective way to get at their understanding. Questionnaires are an evaluation instrument that probes the users about capabilities they expect on each of the devices or components in the ecology.

19.5.6 UX Evaluation Within the Interaction Level of the Pyramid of Needs

Essentially everything in the UX evaluation chapters (Chapters 19–25) applies to the interaction level. The late funnel is best for evaluating within the interaction layer, where you address user actions in task-level design. Therefore most task-based evaluations are best here.

19.5.7 UX Evaluation Within the Emotional Level of the Pyramid of Needs

Many topics in this chapter are generally relevant to evaluating the emotional level, too, and those instances are pointed out as such. In addition, there is an entire section (see Section 22.4) devoted to specific ways to evaluate the emotional level. Qualitative interviews are best here.

19.6 ADAPT AND APPROPRIATE METHODS

19.6.1 You Can Apply Any Given Evaluation Method Over a Range of Rigor

Each UX evaluation method has its own range of potential rigor. For example, you can perform any method in a highly rigorous way that will maximize effectiveness and minimize the risk of errors, subject to limitations innate to the method.

When high rigor is not required, especially in agile UX environments, you can also perform any evaluation method rapidly, with many shortcuts. By filtering and abstracting the evaluation data down to the most important points, you can gain efficiency (higher speed and lower cost).

High rigor is not always a goal. In many design situations, such as early project stages where things are changing rapidly, rigor is not a priority. It is more important to be agile, to iterate and learn quickly. That is why some methods were invented to favor rapidness over rigor, methods specifically designed to be rapid and happily accepting the correspondingly limited rigor.

19.6.2 Adapt and Appropriate UX Evaluation Methods

It is not enough to choose the standard UX evaluation methods and techniques that fit your goals and constraints. You will find it very natural to tailor and tune the methods and techniques to the nuances of your team and project as you appropriate them as your own for each different design situation. This aspect of choosing methods and techniques is discussed at length in Section 2.4. Here we list some design situations and related concerns about choosing appropriate evaluation methods.

- Working on a domain-complex system (see Section 3.2.1.3): Analytic evaluations work best in evaluating the appropriateness of the high-level design choices for that domain because the evaluation is done by experts who are familiar with the domain and with the different market competitors in that domain. Empirical evaluations for these systems work best in evaluating the design choices at a more granular level to ensure they match expectations and mental models of the users who work in this domain. Empirical methods are particularly effective in evaluating systems that should support novice users of domain-complex systems.
- Working on an interaction-complex system (see Section 3.2.1.2): Choice of analytic or empirical approaches comes down to other factors, such as resources available and timelines. Analytic methods employed by teams with deeper UX expertise are cheaper and faster compared to empirical versions. Empirical methods, on the other hand, tend to identify larger number of (and often less severe) problems, useful in polishing the final rough edges in the design.
- Working on a system with low-risk tolerance (e.g., critical systems, such as health care or air traffic control): Although analytic approaches are useful during early stages of the development lifecycle, more rigorous empirical evaluation variants conducted

over multiple versions of the evolving design are appropriate. The goal here is to gain confidence that the design meets or exceeds the stringent UX targets (e.g., no errors in a health care system).

- Working with a team with low UX expertise or maturity: Empirical methods are most appropriate because the effectiveness of analytic methods depends on the expertise of the people doing the evaluation. As an added advantage, observing users interact with designs in empirical evaluations helps understand tradeoffs and problems with the different elements in the design and is often a good way to raise the expertise of the team.
- Working on a novel or cutting-edge system (e.g., a newly introduced system, such as Apple Vision Pro with new interaction paradigms): Analytic methods in the hands of experts help to evaluate the conceptual design and the philosophy of the design and its appropriateness to the system. Empirical methods are then useful in validating the conceptual designs and in identifying other lower-severity problems that are sometimes missed by expert evaluations.

For UX evaluation, as perhaps for most UX work, our motto echoes an old military imperative: Improvise, adapt, and overcome! Be flexible and customize your methods and techniques, creating variations to fit your evaluation goals and needs. This includes adapting any method by leaving out steps, adding new steps, and changing the details of a step.

19.7 SOME BACKGROUND

19.7.1 An Axe Analogy to Explain the Difference Between Empirical and Analytic Evaluation

In describing the distinction between experienced or observed usage of something (empirical) vs. examination of it (analytic), Scriven (1967) proposed an oft-quoted (Carroll, Singley, & Rosson, 1992; Gray & Salzman, 1998) analogy featuring an axe: "If you want to evaluate a tool, say an axe, you might study the design of the bit, the weight distribution, the steel alloy used, the grade of hickory in the handle, etc. [intrinsic or analytic], or you might just study the kind and speed of the cuts it makes in the hands of a good axeman [payoff or empirical]" (p. 53 [text in brackets added]). In Hartson and colleagues (2003), we added our own embellishments to this comparison.

Although this example served Scriven's (1967) purpose well, it also offers us a chance to make a point about the need to identify UX goals carefully before establishing evaluation criteria. Giving a UX perspective to the axe example, we note that user performance observation in payoff (empirical) evaluation does not necessarily require a good axeman (or axeperson). UX goals (and therefore evaluation goals) depend on expected user classes of key work roles and the expected kind of usage. For example, an axe design that gives optimum performance in the hands of an expert may be too dangerous for a novice user. For the inexperienced user, safety may be a UX goal that transcends firewood production.

19.7.2 Concerns About UX Evaluation Methods

19.7.2.1 Measurability of user experience parameters: A problem on the empirical quantitative side

Quantitative evaluation of an attribute, such as usability or UX, implies some kind of measurement. But, perhaps surprisingly, neither usability nor user experience is directly measurable. Usability is an abstract characteristic of a design, and user experience is something felt internally by the user. In fact, most interesting phenomena, such as teaching and learning, share the same difficulty. So, we resort to measuring things we can measure and using those measurements as indicators of our more abstract and less measurable notions. For example, we can understand usability effects, such as productivity or ease of use, by measuring observable user performance-based indicators, such as time to complete a task, and counts of errors encountered by users within task performance.

19.7.2.2 Reliability of formative UX evaluation methods: A problem on the qualitative side

In simple terms, the reliability of a formative UX evaluation method means repeatability, and it applies to both empirical and analytic methods (Hartson, Andre, & Williges, 2003). Poor reliability means that, although you use the same method applied to the same evaluation target, you get different results (lists of UX problems) across different user participants (for empirical methods) or across several different UX experts or inspectors (for analytic methods). In practice, regardless of the method you use—even thorough lab-based empirical methods—the differences can be fairly large, and this has been a cause for alarm in some academic and theoretical circles.

Why so much variation? UX evaluation is difficult. In our opinion, the biggest reason for the limitations of our current methods is that evaluating UX, especially in large system designs, is very difficult. No one has the resources to look everywhere and test every possible feature on every possible screen or webpage associated with every possible task. You will not find all the UX problems in all those places. One evaluator may find a problem in a place that other evaluators did not examine. So, it cannot be surprising that evaluators come up with comprehensive problem lists that are unique to each evaluator.

Individual differences naturally cause variations in results. The literature contains many discussions and debates about the effectiveness and reliability of various UX evaluation methods. Effectiveness across the field of UX evaluation has been called into question. We cannot hope to replicate that level of discussion here, but we do offer a few insights into the problem.

When variation of UX evaluation results is due to using different evaluators, it is called the evaluator effect (Hertzum & Jacobsen, 2003; Vermeeren, van Kesteren, & Bekker, 2003). Different people see usage and problems differently and have different problem detection rates. They naturally see different UX problems in the same design. Different evaluators even report very different problems when observing the same evaluation session. Even the same person can get different results in two successive evaluations of the same system, and different UX teams interpret the same evaluation report in different ways.

Rapid UX evaluation (discount) methods. The situation may appear even worse when we introduce less rigorous methods. A rapid UX evaluation method results from trading off the thoroughness of high rigor for low cost and fast application. Rapid methods include most of the analytical data collection methods, especially in their less rigorous forms. Because these inspection techniques are less costly, they have been called discount evaluation techniques (Nielsen, 1989).

Criticism of discount methods. Rapid UX evaluation methods, especially inspection methods, have been criticized as "damaged merchandise" (Gray & Salzman, 1998) or "discount goods" (Cockton & Woolrych, 2002). Use of rapid was intended as a positive reflection of the advantage of fast turnaround and lower costs; use of damaged and discount was pejoratively intended to connote inferior, bargain-basement goods because of reduced effectiveness and susceptibility to errors in identifying UX problems.

Real limitations. The major downside of inspection methods, which employ experts rather than real users, is they suffer from reliability problems (Section 19.7.2.2) and tend to find a higher proportion of lower-severity problems. They can also suffer from validity problems, yielding false positives (occurrences identified as UX issues but that turn out not to be real or less severe problems) and missing some UX problems because of false negatives.

Another potential drawback is that the UX experts doing the inspection may not know the subject-matter domain or the system in depth. This can lead to a less effective inspection, but it can be offset by including a subject-matter expert on the inspection team.

It may be some comfort, however, to know that these weaknesses are inherently in all kinds of UX evaluation, including our venerable yardstick of performance, the rigorous lab-based empirical formative evaluation (Molich, Bevan, & Butler et al., 1998; Molich, Thomsen, & Karyukina et al., 1999; Newman, 1998; Spool & Schroeder, 2001), because many of the phenomena and principles are the same. Formative evaluation, in general, is not very reliable or repeatable, yet it works well enough to be worthwhile.

The good news. Even UX evaluation methods with low reliability can be effective—that is, the methods find UX problems that need fixing, and often they find the most important problems (Hartson, Andre, & Williges, 2003). Now, rapid and low-cost UX evaluation methods are the de facto standard, and they are used effectively every day in agile UX practice. The bulk of UX evaluation practice is about learning as much as you can about problems with the design, using the resources you have, then moving on from there.

Be practical and make it good enough. Wixon (2003) spoke up for the practitioner in the discussion of usability evaluation methods. From an applied perspective, he pointed out that a focus on validity and proper statistical analysis in these studies does not serve the needs of practitioners in finding the most suitable usability evaluation method and best practices for their work context in the business world.

Discount methods are the way forward. In sum, although criticized as false economy in the academic HCI literature, discount methods are practiced heavily and successfully in the field. Among reasons for success include the following:

- Iteration helps close the gap.
- Because evaluators have individual differences, adding evaluators helps because one evaluator may detect a problem that others did not.
- We are using our experience as UX practitioners and not doing the process with our eyes closed.
- Perhaps most importantly, we have matured as a discipline, having established well-tested design patterns and design libraries that help ensure a minimum level of UX quality.

In the hands of an experienced evaluator, inspection methods can be relatively fast and very effective. You can get a significant portion of UX problems dealt with and out of the way at a low cost. Under the right conditions, you can do a UX inspection and its analysis, fix the major problems, and update the prototype design in the same day! There are so many variables that, to be practical, we have to live with imperfect reliability.

19.8 SEGUE

In this chapter, we established foundational concepts of UX evaluation. In the next chapter, we look at goals, metrics, and targets for empirical UX evaluation.